Inference for logistic regression models

Agenda

- For homework submissions:
  - Make sure to submit rendered HTML
  - If writing math in Quarto (e.g. regression models, maximum likelihood problems, etc.), make sure the equations display correctly
- Discuss feedback survey
- Recap activity from end of last class
- More inference!

Last time: Activity

$$Admit_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 GPA_i + \beta_2 GRE_i + \beta_3 Rank2_i + \beta_4 Rank3_i + \beta_5 Rank4_i$$

> We want to test whether there is a relationship between Rank and grad school acceptance, after accounting for GPA and GRE. What are my null and alternative hypotheses?

H_0: $\beta_3 = \beta_4 = \beta_5 = 0$

H_A: at least one of $\beta_3, \beta_4, \beta_5 \neq 0$

Activity

$$Admit_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 GPA_i + \beta_2 GRE_i + \beta_3 Rank2_i + \beta_4 Rank3_i + \beta_5 Rank4_i$$

$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$

H_A : at least one of $\beta_3, \beta_4, \beta_5 \neq 0$

> What is the reduced model?

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 GPA_i + \beta_2 GRE_i$$

Activity

$G =$ deviance$_{\text{reduced}}$ $-$ deviance$_{\text{full}}$

$= 480.3 - 458.5 = 21.8$

```
##
## Call:  glm(formula = admit ~ gpa + gre + factor(rank), family = binomial,
##     data = grad_app)
##
## Coefficients:
##   (Intercept)            gpa            gre  factor(rank)2  factor(rank)3
##     -3.989979       0.804038       0.002264      -0.675443      -1.340204
## factor(rank)4
##     -1.551464
##
## Degrees of Freedom: 399 Total (i.e. Null);  394 Residual
## Null Deviance:        500
## Residual Deviance: 458.5      AIC: 470.5
```

← deviance for intercept-only model

$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0$

```
##
## Call:  glm(formula = admit ~ gpa + gre, family = binomial, data = grad_app
##
## Coefficients:
## (Intercept)          gpa          gre
##   -4.949378     0.754687     0.002691
##
## Degrees of Freedom: 399 Total (i.e. Null);  397 Residual
## Null Deviance:        500
## Residual Deviance: 480.3      AIC: 486.3
```

Activity

$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$

H_A : at least one of $\beta_3, \beta_4, \beta_5 \neq 0$

Drop-in-deviance test: $G = 21.8$

> What is the distribution of G under the null hypothesis?

Under H_0, $G \sim \chi^2_3 \leftarrow$ testing 3 parameters ($\beta_3, \beta_4, \beta_5$)

Activity

$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$

H_A : at least one of $\beta_3, \beta_4, \beta_5 \neq 0$

Drop-in-deviance test: $G = 21.8$

Under $H_0, G \sim \chi^2_3$

```r
pchisq(21.8, df=3, lower.tail=F)
```

```
## [1] 7.178958e-05
```

A new question

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 GRE_i$$

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.901344   0.606038  -4.787 1.69e-06
gre          0.003582   0.000986   3.633  0.00028
```

Here $\widehat{\beta}_1 = 0.003582$. Will $\widehat{\beta}_1$ be exactly equal to the true value β_1?

A new question

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 GRE_i$$

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            Estimate Std. Error z value Pr(>|z|)
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gre          0.003582   0.000986   3.633  0.00028
```

> Here $\widehat{\beta}_1 = 0.003582$. Will $\widehat{\beta}_1$ be exactly equal to the true value β_1?

No -- our estimate comes from one sample of data, and there is variability from sample to sample.

> How can I account for this variability when reporting an estimate of β_1?

Interval estimates

Point estimate: a single "best guess" value of a parameter. The estimated slope $\widehat{\beta}_1$ is a point estimate of β_1

Interval estimate: a "reasonable range" of possible values for a parameter. A *confidence interval* is an interval estimate

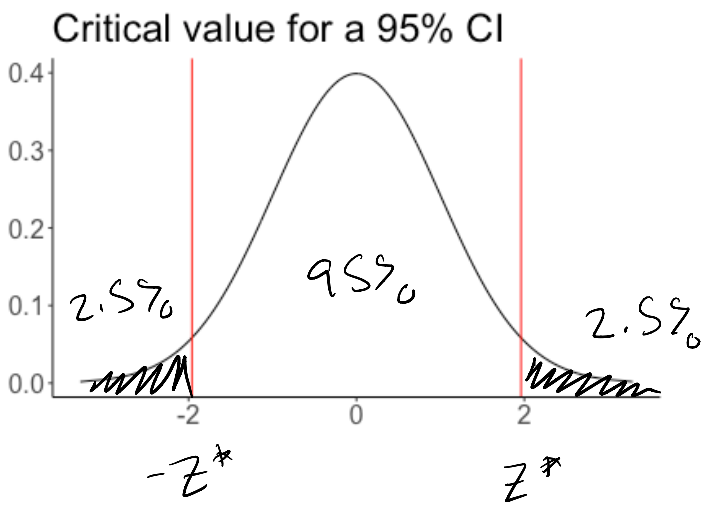
> How would I construct a confidence interval for β_1?

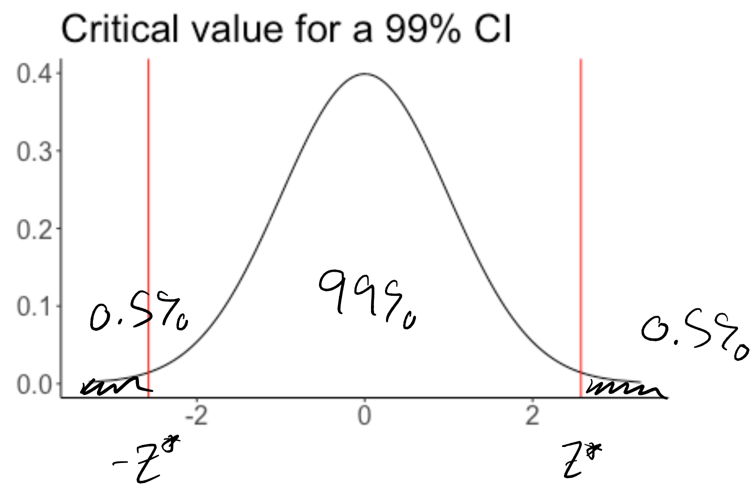
$$\hat{\beta}_1 \pm \underbrace{z^*}_{\text{critical value}} SE(\hat{\beta}_1)$$

(how many standard deviations do I look in each direction)

The critical value z^*

To construct a $(1-\alpha)\cdot 100\%$ confidence interval, z^* is the value such that $P(Z > z^*) = \alpha/2$





95% CI :

$(1-0.05)\cdot 100\%$

$\alpha = 0,05$

$\frac{\alpha}{2} = 0.025$

Computing z^*

Example: for a 95% confidence interval, z^* = 1.96

look in right tail

```
qnorm(0.025, lower.tail=FALSE)
```

quantile of a normal distribution

```
## [1] 1.959964
```

Example: for a 99% confidence interval, z^* = 2.58

```
qnorm(0.005, lower.tail=FALSE)
```

```
## [1] 2.575829
```

Confidence interval

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-2.901344	0.606038	-4.787	1.69e-06
gre	0.003582	0.000986	3.633	0.00028

95% confidence interval for β_1:

$$\hat{\beta}_1 \pm z^* SE(\hat{\beta}_1)$$

$$= 0.0036 \pm 1.96(0.00099)$$

Motivating the confidence interval

- $\widehat{\beta}_1$ is our "best guess" of β_1
- But, there is variability in our estimate of β_1; $\widehat{\beta}_1$ will vary from sample to sample

Idea: What if we take a "new" sample of data, and see how $\widehat{\beta}_1$ changes?

Bootstrap sample

A **bootstrap sample** is a sample of rows from the data, *with* replacement

Original data:

GRE	Admit
380	0
~~660~~	~~1~~
800	1
520	0
760	1
~~640~~	~~1~~

Bootstrap sample:

GRE	Admit
380	0
380	0
800	1
520	0
760	1
760	1

Motivation: Variability in resampling from the *observed* data should reflect variability in resampling from the *population*

Bootstrap sample

A **bootstrap sample** is a sample of rows from the data, *with* replacement

Original data:

GRE	Admit
380	0
~~660~~	~~1~~
800	1
520	0
760	1
~~640~~	~~1~~

Another bootstrap sample:

GRE	Admit
380	0
760	1
520	0
800	1
520	0
800	1

Motivation: Variability in resampling from the *observed* data should reflect variability in resampling from the *population*

Activity

- Take many bootstrap samples
- For each bootstrap sample, calculate $\widehat{\beta}_1$
- Examine the distribution of estimated slopes

https://sta214-f25.github.io/class_activities/ca_15.html

Activity

```r
nsim <- 1000
boot_ests <- rep(NA, nsim)

for(i in 1:nsim){
  grad_boot <- grad_app |>
    slice_sample(n = nrow(grad_app), replace=T)

  m_boot <- glm(admit ~ gre, family = binomial,
                data = grad_boot)
  boot_ests[i] <- coef(m_boot)[2]
}
```

Activity

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.901344   0.606038  -4.787 1.69e-06
gre          0.003582   0.000986   3.633  0.00028
```

```
mean(boot_ests)
```

```
## [1] 0.003555198
```

≈ 0.003582 $\quad \hat{\beta}_1$

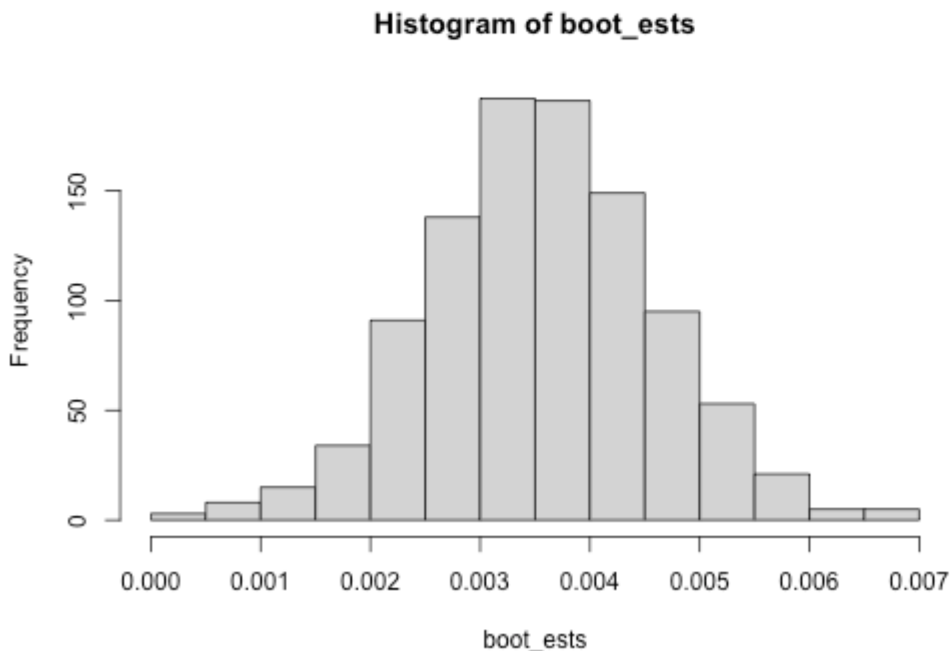
```
sd(boot_ests)
```

```
## [1] 0.001040903
```

≈ 0.000986 $\quad SE(\hat{\beta}_1)$

> How do the mean and standard deviation of the bootstrap estimates compare to the regression output?

Activity

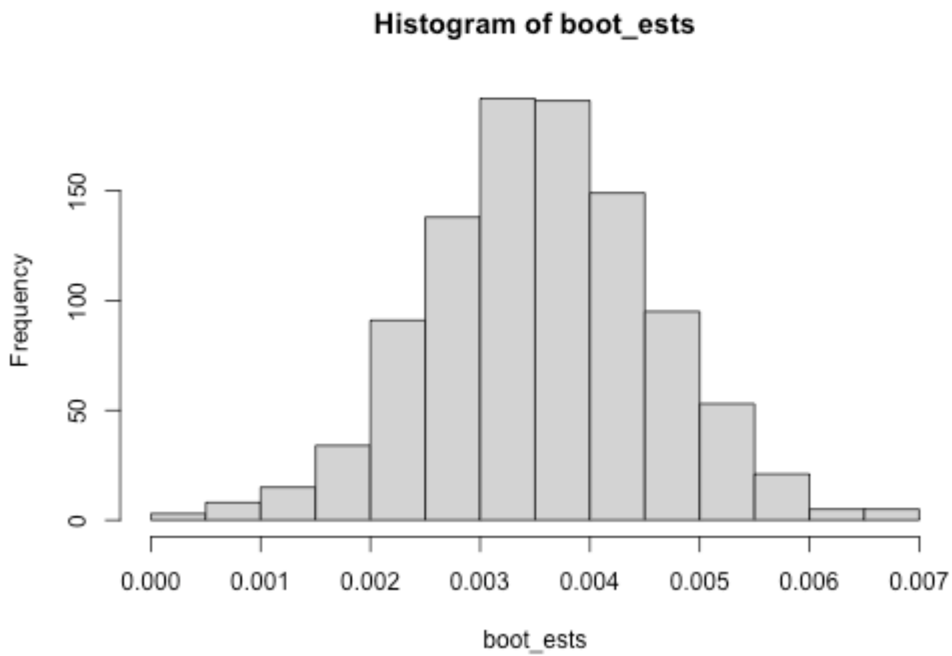


What does the bootstrap distribution look like?

Normal, centered at $\hat{\beta}_1$ from original data

std dev = $SE(\hat{\beta}_1)$ from original data

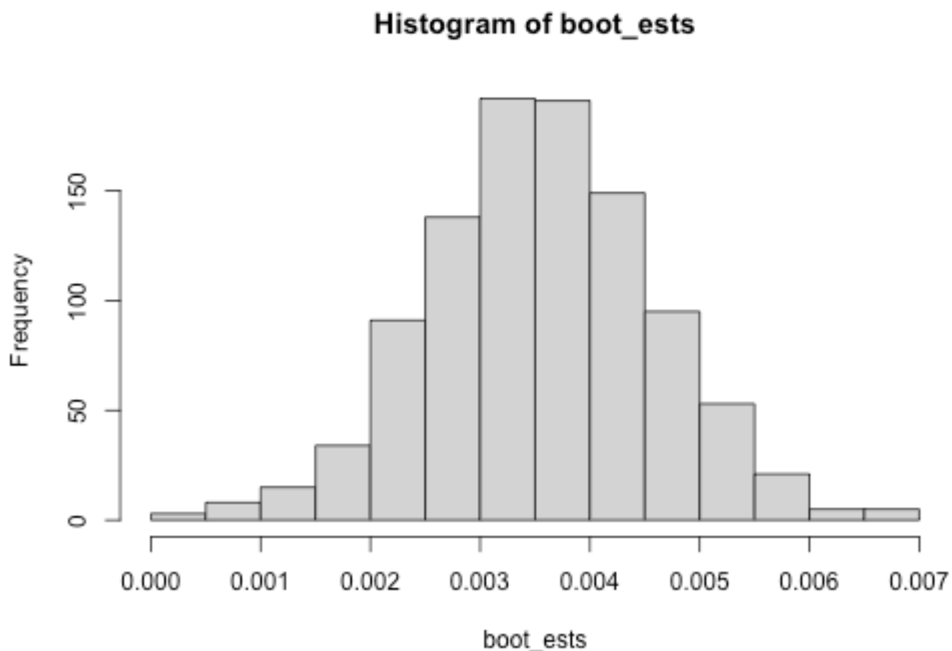
Activity



Based on the bootstrap distribution, is a value of 0.004 a "reasonable" guess for β_1?

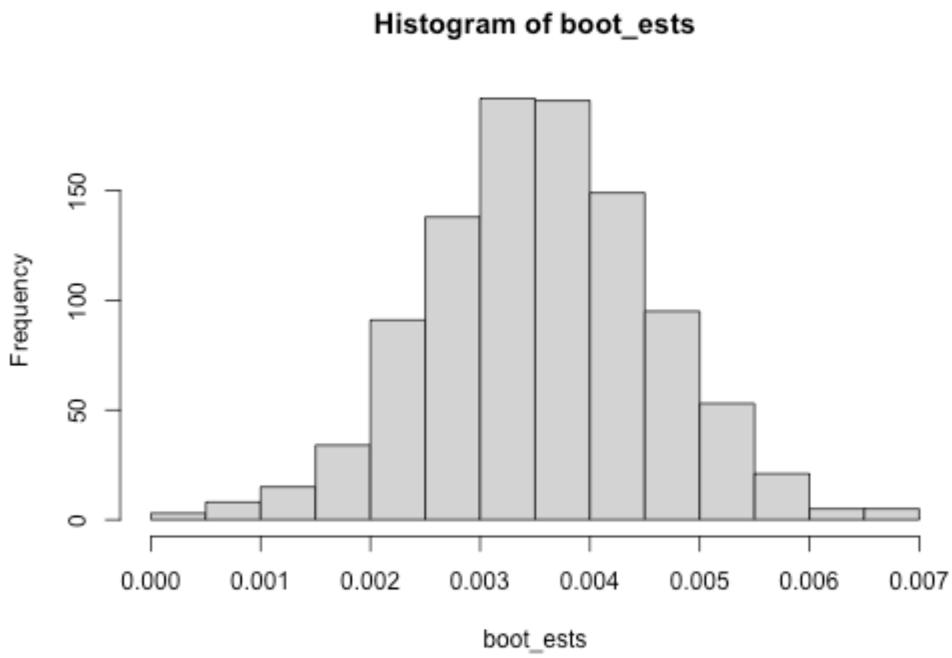
reasonable

Activity



Based on the bootstrap distribution, is a value of 0.0025 a "reasonable" guess for β_1? reasonable

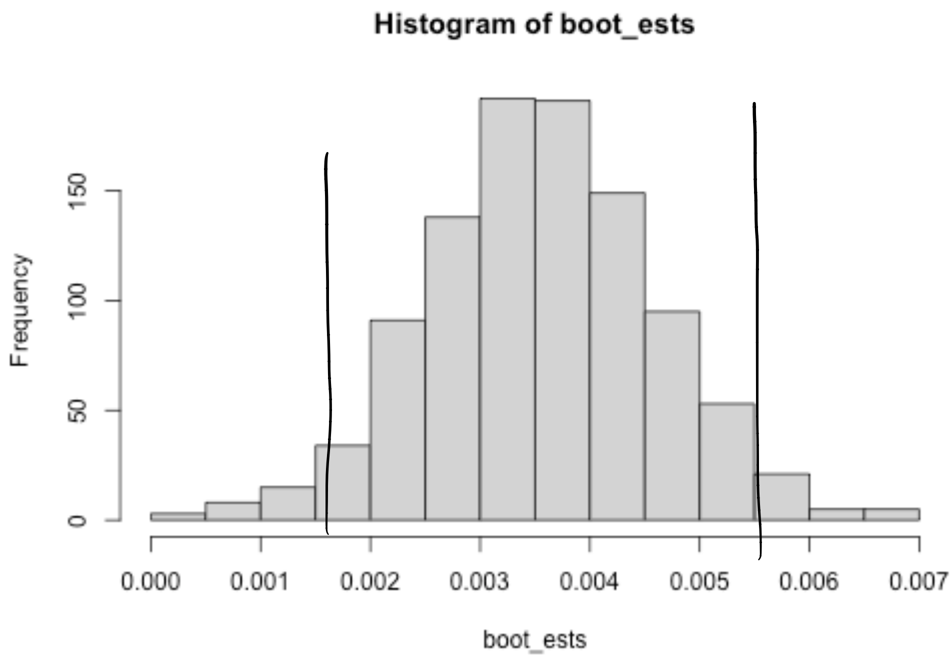
Activity



> Based on the bootstrap distribution, is a value of 0.1 a "reasonable" guess for β_1?

unreasonable

Activity

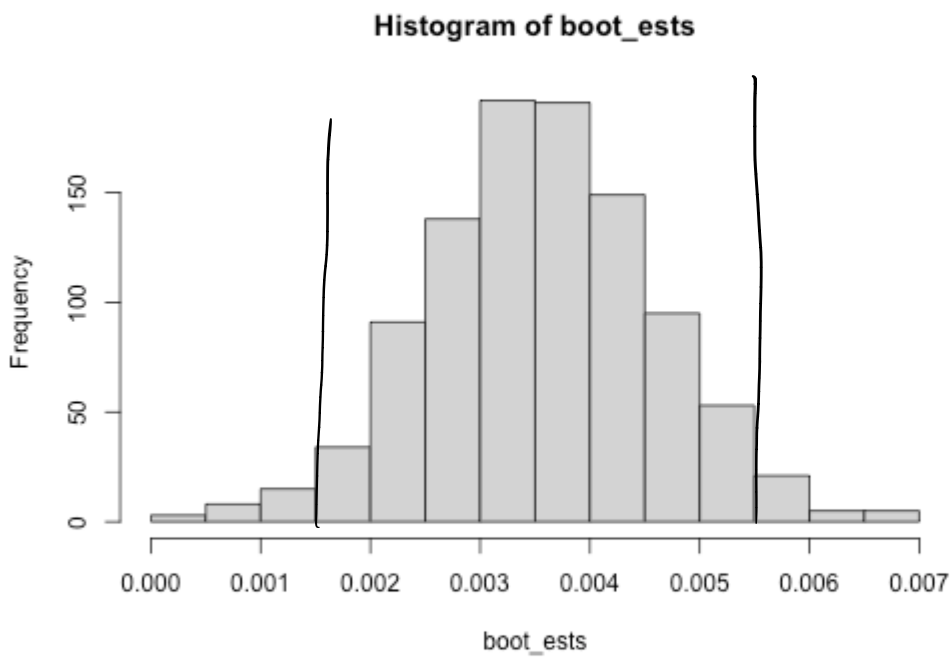


> Based on the bootstrap distribution, what would be a "reasonable range" of values for β_1?

Confidence interval

95% confidence interval for β_1:

$$\widehat{\beta}_1 \pm 1.96 SE(\widehat{\beta}_1) = 0.00358 \pm 1.96(0.00099) = (0.0016, 0.0055)$$



Activity

Work on the activity (handout).

Activity

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.73743    0.08499   20.44   <2e-16 ***
WBC         -0.36085    0.01243  -29.03   <2e-16 ***
```

> Calculate a 99% confidence interval for the change in the log odds of dengue associated with a one-unit increase in WBC.

Activity

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.73743    0.08499   20.44   <2e-16 ***
WBC         -0.36085    0.01243  -29.03   <2e-16 ***
```

> How could we calculate a 99% confidence interval for the change in the *odds* of dengue associated with a one-unit increase in WBC?